PRODUCT PERFORMACE UPDATE

25% - 29% Extended Life of EAF Electrodes at Gerdau Monroe

ELECTRODE DETAILS

- **Electrode Consumption:** Current consumption rates averaged between 3 – 4 lbs/per ton on 24" diameter electrodes. Average heat times are between 42 to 45 minutes in length with an average tap temperature of 3050° F. Material produced is carbon steel.

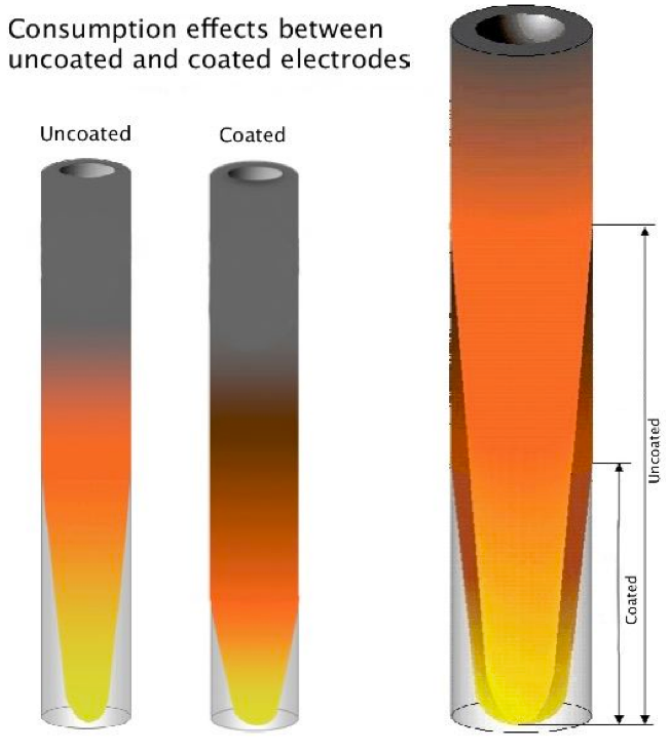
- **Customer Need:** ITC recommended a fully automated spraying solution to reduce electrode consumption rates and lower maintenance/delay time associated with electrode additions.

ITC RECOMMENDATION

- **Engineered Approach:** ITC recommended their recently developed EC 303 ceramic coating specifically design for electrodes, ITC 303 EC, which is formulated for application on carbon/graphite substrates and can be sprayed while the substrate is at elevated temperatures up too 900° F.

- **High Emissivity Coatings:** ITC 303 EC is a proprietary coating designed to work in the most extreme conditions. By using ITC 303 EC, the following benefits were achieved;

  - Reduction of carbon/graphite consumption, on average, by 25% to 30%
    - Reduced side oxidation
      1) Reduced oxidation cone
      2) Increased tip diameter
    - Reduced tip consumption
      1) Reduced number of electrode connections
      2) Reduced tip loss
  - Improved conductivity of electrodes
    - By redistribution of the electrical current flow
  - Additional heats between electrode additions
    - With longer electrode life, additional heats per day or month are realized
  - Time savings
    - Improved working conditions for furnace operators and staff
    - Reduced crane time



IMPACT ON THE PROCESS

- REDUCTION IN ELECTRODE CONSUMPTION
  The use of the coating significantly lowered consumption rates by 25% to 30%.

- REDUCED MAINTENANCE & DOWNTIME/DELAY
  With increased electrode life, maintenance/crane time was reduced due to less electrode additions.

- IMPROVED CONDUCTIVITY OF ELECTRODES
  With use of the coating, electrical current flow is now channeled out through the electrode tip and all stray electric arcs are eliminated potentially increasing melt rate and lowering melt times.

- INCREASE IN NUMBER OF HEATS PER MONTH
  Having less delays required for electrode additions and longer electrode life, additional time captured could result in one or several additional heats per month, increasing total yield.